



Model Context Protocol (MCP)

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What Is MCP?

- Open standard (JSON-RPC 2.0) introduced by Anthropic Nov 2024.
- Defines how an AI client requests context from a server (uses a tool)
- Goal: One protocol, many tools—swap LLM vendors or data sources without new glue.

Why We Need MCP

- LLM apps traditionally require bespoke connectors for every API, DB, or SaaS tool.
- Integrations become brittle and hard to audit as feature counts grow.
- Teams lose time reinventing work flows, error handling, and data validation.

Core Components

Role	Lives In	Purpose
MCP Client	Host app / agent	Packages model calls into MCP messages
MCP Server	External service wrapper	Advertises Capabilities
Capabilities	Tools · Resources · Prompts	Declarative metadata + JSON Schema

Request/Response Flow



The Magic: How LLMs decides which tool

- System Prompting with Tools List:
 - LLM is informed about available tools at conversation start.
 - Each tool has a name, description, and input/output schema.
- Internal Planning (Chain-of-Thought)
 - Model reasons if it needs external help: "I can't guess weather — I must call `get_weather`."

A Sample Tool Call

- LLM outputs a special tool_call **Token**
- Formatted in json to reduce hallucination

```
{  
  "tool_call": {  
    "name": "get_weather",  
    "arguments": { "location": "Paris" }  
  }  
}
```

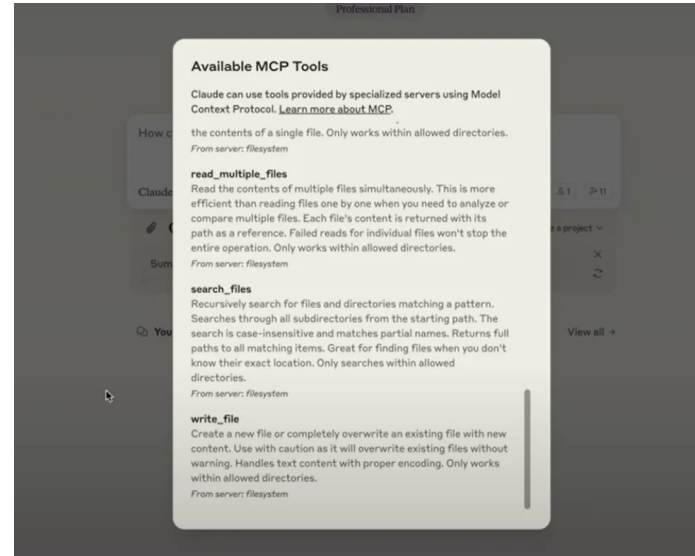
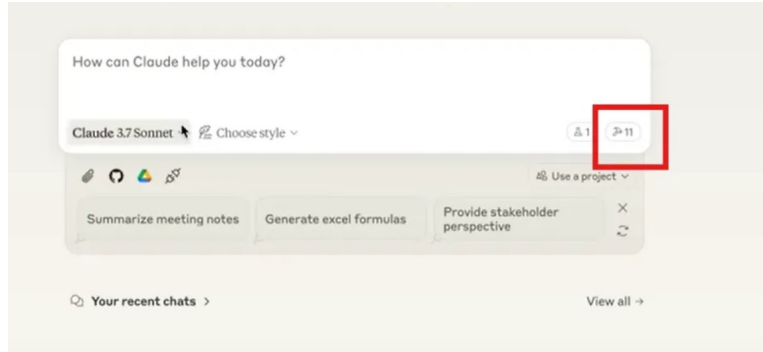
Real-World Example: Organizing Files with Claude MCP

If we have messy project folder: .py, .txt, .md, .csv files.

User wants Claude to organize local file folders

1. Download Claude desktop which has mcp feature
2. Connect to a local file MCP server using this configuration:

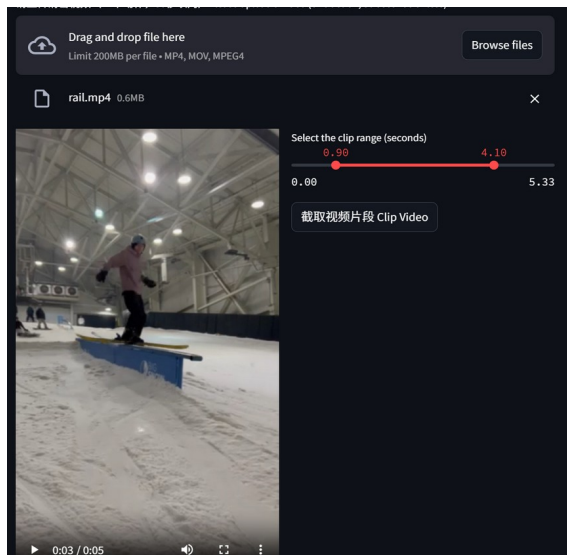
```
{
  "servers": [
    {
      "name": "filesystem",
      "command": "npx @modelcontextprotocol/server-fileSystem",
      "args": ["/Users/username/Documents", "/Users/username/Downloads"]
    }
  ]
}
```

Now we can ask claude: "Organize files: scripts to /scripts, documents to /docs, data to /data."

A little AD: AI Snow Coach

<https://snowcoach.streamlit.app>



What Went Well:

- **Approach and Speed:** The skier maintains a controlled and straight approach toward the rail, which is essential for a successful rail slide. The speed is moderate and not overly aggressive, decreasing the risk of losing balance.
- **Commitment:** The athlete fully commits to getting onto the rail. Hesitation is a common issue for beginners, but this skier confidently pops onto the feature.
- **Upper Body Position:** The skier keeps their arms out for balance, providing stability during the movement.
- **Eyes on the End:** Throughout the approach and while sliding the rail, the skier's head appears to be looking toward the end of the rail, which is crucial for maintaining direction and balance.

Areas for Improvement:

- **Pop and Centering:** The pop onto the rail seems insufficiently vertical—the skier comes on slightly from the side rather than directly above, which can unbalance the slide.
- **Weight Distribution:** The weight appears to shift toward the tails of the skis, causing the body to lean back at certain frames. This not only reduces rail control but also increases the risk of slipping out.
- **Hip and Shoulder Alignment:** There is rotation in the shoulders, which causes the lower body to follow and increases the chance of spinning out or coming off the rail unintentionally. Alignment should be more perpendicular to the rail.
- **Knees & Ankles Flexion:** The skier could bend the knees and ankles a bit more to better absorb any rail vibrations and maintain lower center of gravity.

One Critical Improvement: The most critical improvement for this skier is weight distribution. Throughout the slide, more of the skier's weight is on their heels/tails, causing the body to lean back rather than staying centered over the rail. This makes it hard to maintain balance, control, and to absorb imperfections in the surface. The skier should focus on keeping their weight centered, with shins pressing into the front of the boots and hips stacked over the feet to stay balanced and secure throughout the slide.

References

- [https://sebastian-petrus.medium.com/mcp-server-
claude-work-with-local-files-594f332a7084](https://sebastian-petrus.medium.com/mcp-server-
claude-work-with-local-files-594f332a7084)